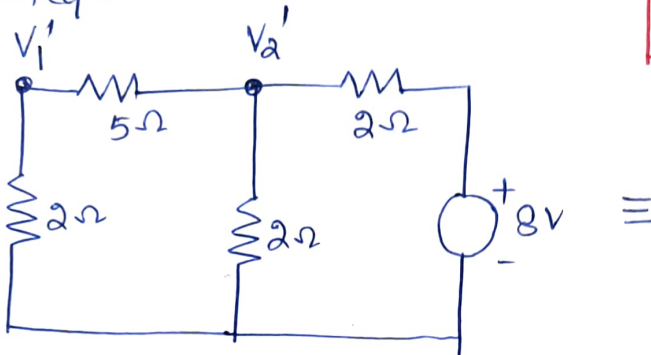
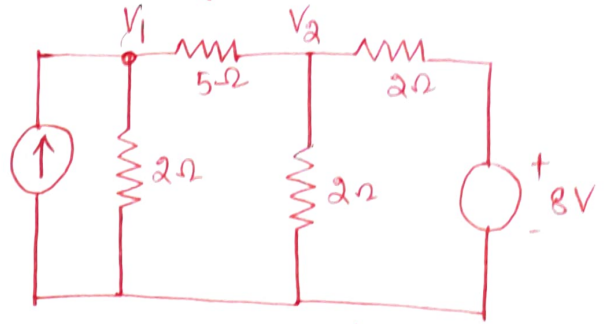


Q. for the circuit shown find the node voltages V_1 and V_2 using superposition theorem.

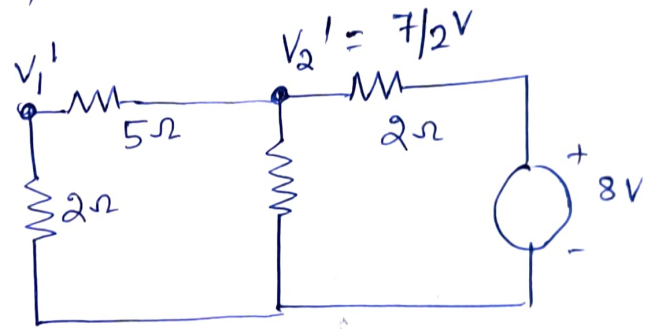
STEP:1: voltage source is Applied and current source is open circuited -



$$R_{eq} = (2+5) \parallel 2 = 7 \parallel 2 = 14/9 \Omega$$

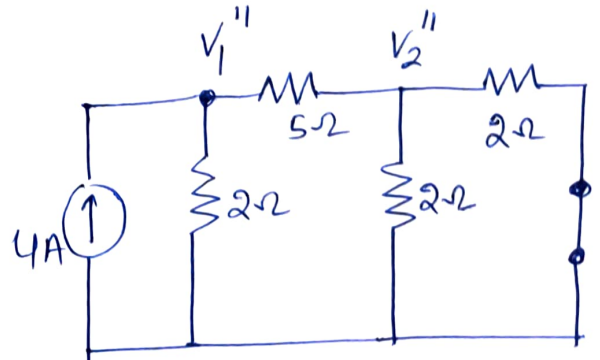
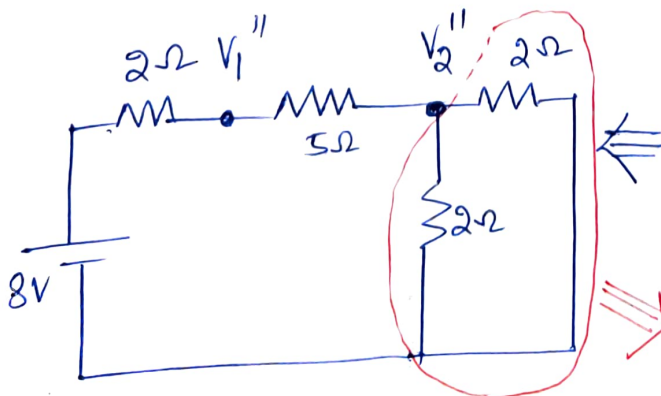
$$V_2' = \frac{8 \times 14/9}{14/9 + 2} = \frac{-8 \times 14}{-32/9} = 7/2 \text{ V}$$

$$V_1' = \frac{V_2' \times 2}{5+2} = \frac{7/2 \times 2}{7} = 1 \text{ V}$$

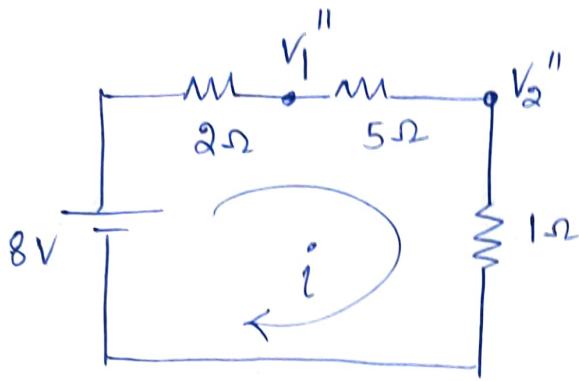


$$V_2' = 7/2 \text{ V} ; V_1' = 1 \text{ V}$$

STEP:2: only current source '4A' is considered and voltage source is short cktd.



$$R_{eq} = 1 \Omega$$



$$i = \frac{8V}{8\Omega} = 1A$$

$$V_2'' = 1A \times 1\Omega$$

$$V_2'' = 1V$$

$$V_1'' = 8 - 2 \times i$$

$$V_1'' = 8 - 2 \times 1 = 6V$$

$$V_2 = V_2' + V_2'' = \frac{7}{2} + 1 = \frac{9}{2}V$$

$$V_1 = V_1' + V_1'' = 1 + 6 = 7V$$